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Satellite Data Visualization Platform for Weather and Climate Pattern Analysis

A CASE STUDY ANALYSIS REPORT

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1. Scenario Study & Domain Analysis

Business Domain

The platform operates within the Geospatial & Climate Informatics (Earth Observation / GovTech) domain. It bridges the gap between raw scientific remote-sensing feeds (NASA, NOAA, ESA) and downstream decision-makers (disaster responders, planners, policy teams, researchers).

Key Stakeholders

- **Meteorologists & Climate Researchers:** Require raw-to-processed conversion, complex analytical overlays, time-series analysis, and export features.
- **Disaster Management Teams & First Responders:** Require low-latency real-time visualizations, severe event warnings, and simplified risk maps.
- **Agricultural & Urban Planners:** Require localized trend analysis (vegetation health, precipitation anomalies, surface temperature shifts).
- **Non-Expert Users / Policy Makers / Students:** Require intuitive point-and-click maps, pre-rendered layer toggles, and plain-language summaries.
- **System Administrators & Data Engineers:** Responsible for pipeline maintenance, API integration, cloud infrastructure, and ingestion status monitoring.

Existing Challenges

- **Data Friction:** Data is locked in complex formats (HDF5, NetCDF, GRIB2, GeoTIFF) requiring specialized software like QGIS, ArcGIS, or Python (xarray, rasterio).
- **Latency & Overhead:** Manual ETL (Extract, Transform, Load) pipelines delay critical decision-making during fast-evolving weather events (e.g., cyclone tracking).
- **Lack of Interoperability:** Cross-referencing disparate indicators (e.g., NDVI vs. soil moisture vs. sea surface temperature) requires tedious manual alignment.

System Vision

An open-access, web-native geospatial analytics platform that automatically ingests raw satellite raster/vector feeds, standardizes them into cloud-optimized formats (e.g.,

Cloud-Optimized GeoTIFF / Zarr), and delivers fast, multi-layer web map rendering alongside automated anomaly detection.

2. Functional & Non-Functional Requirements

Functional Requirements (FR)

ID	Category	Requirement Description
FR-01	Data Ingestion	System shall continuously ingest real-time and historical satellite feeds from public APIs (NASA Earthdata, NOAA NCEI, Copernicus Open Access Hub).
FR-02	Data Processing	System shall convert incoming raw raster formats (HDF5, NetCDF) into Cloud-Optimized GeoTIFF (COG) or Zarr tiles for web delivery.
FR-03	Map Visualization	System shall provide an interactive 2D/3D map canvas supporting layer toggling, opacity adjustment, time-slider scrubbing, and spatial zooming.
FR-04	Time-Series Analysis	System shall generate dynamic charts (line graphs, histograms) for selected point locations or user-drawn Bounding Boxes (AOIs) across custom date ranges.
FR-05	Anomaly Detection	System shall run statistical algorithms (e.g., Z-Score, baseline delta comparison) to flag temperature spikes, cyclone formations, and rainfall deficits
FR-06	Multi-Layer Overlay	System shall allow users to stack and blend multiple environmental layers (e.g.,

		Precipitation + NDVI) with custom color maps.
FR-07	Region Filtering	System shall filter data using administrative boundaries (country, state, district) or custom uploaded GeoJSON/KML spatial polygons.
FR-08	Data & Map Export	System shall support exporting visual maps as high-resolution PDF/PNG images, and underlying clipped datasets as GeoTIFF, CSV, or GeoJSON.

Non-Functional Requirements (NFR)

- **Performance:**
 - Map tile rendering response time must be < 1.5 seconds under standard broadband conditions.
 - Time-series chart generation for up to a 5-year historical window must complete within < 3.0 seconds.
- **Scalability:**
 - Horizontal autoscaling to handle up to $10,000$ concurrent active map users during severe weather events.
 - Cloud storage scaling capable of indexing multi-terabyte dataset expansions without degradation in query speed.
- **Usability:**
 - Intuitive UI adhering to WCAG 2.1 Level AA accessibility standards.
 - Zero client-side desktop software installation required; fully browser-native.
- **Reliability & Availability:**
 - 99.9% system uptime for critical visualization endpoints.
 - Automated ingestion retry logic with exponential backoff if remote space agency APIs fail.
- **Security:**
 - HTTPS/TLS 1.3 encryption across all communication channels.
 - Role-Based Access Control (RBAC) for platform administrative panels and API key rate-limiting.

- **Maintainability:**

- Modular microservices architecture separating Data Ingestion, Tile Server, Analytics Engine, and Web UI.

3. Actors & Use Cases

Actors

1. **General Public / Policy Maker (Primary Actor):** Explores pre-rendered maps, views regional warnings, and reads summary metrics without technical setup.
2. **Researcher / Analyst (Primary Actor):** Draws custom AOIs, executes time-series analytics, runs multi-layer comparisons, and exports raw/processed data.
3. **Disaster Manager (Primary Actor):** Monitors real-time risk layers, receives anomaly alerts, and exports high-impact threat maps for emergency routing.
4. **System Administrator (Secondary Actor):** Monitors pipeline health, manages ingestion schedules, handles user access, and updates color palettes/basemaps.
5. **External Satellite APIs (External Actor):** Data suppliers (NASA, NOAA, Copernicus) providing raw telemetric streams.

Core Use Case Index

- **UC-01:** View Real-Time & Historical Satellite Layers
- **UC-02:** Perform Time-Series Trend Analysis
- **UC-03:** Detect & Highlight Climate Anomalies
- **UC-04:** Perform Multi-Layer Spatial Overlay
- **UC-05:** Filter Data by Spatial Region (AOI)
- **UC-06:** Export Visual Maps and Raw Clipped Datasets
- **UC-07:** Manage Automated Data Ingestion Pipelines

Use Case Diagram (PlantUML)

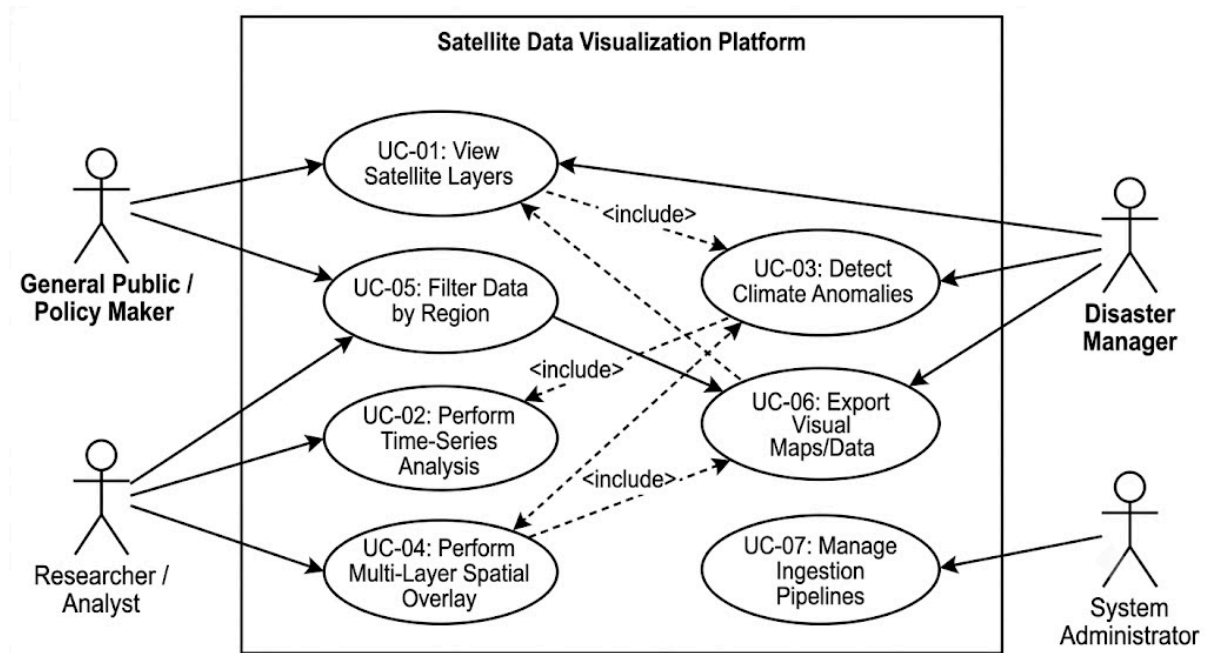


Fig.1:UML Use Case Diagram: Satellite Data Visualization Platform

4. Software Requirement Specification (SRS)

4.1 Introduction

This SRS defines the functional and non-functional specifications for the Satellite Data Visualization Platform for Weather and Climate Pattern Analysis. It outlines data flows, system behavior, interface constraints, and architectural expectations.

4.2 Problem Description

Climate datasets generated by global space agencies are voluminous, highly fragmented, and encoded in specialized binary formats. Non-expert stakeholders lack the tools and computing infrastructure required to process, aggregate, and visualize this information dynamically. This system eliminates that barrier by automating ingestion, transforming formats server-side, and providing an interactive browser interface.

4.3 Scope of the System

- **In Scope:** Automated ingestion of public satellite feeds, conversion to Cloud-Optimized GeoTIFFs/Zarr, interactive web visualization, serverless time-series analysis, statistical anomaly flag generation, multi-layer blending, regional polygon masking, and data downloading.

- **Out of Scope:** Physical satellite command/control, proprietary sensor integration without public API standards, numerical weather prediction model training (e.g., training custom GFS/ECMWF models from scratch).

4.4 Data Requirements

- **Ingestion Inputs:** NetCDF4, HDF5, GeoTIFF, GRIB2 feeds containing variables: Sea Surface Temperature (SST), Land Surface Temperature (LST), Precipitation (TRMM/GPM), Cloud Cover, and NDVI.
- **Internal Storage Schema:** Metadata stored in PostgreSQL/PostGIS (spatial metadata, ingestion logs, regional boundary vectors). Raster grids stored in Object Storage (S3/MinIO) as COGs.
- **Tile Serving Schema:** Dynamic vector and raster tiles generated via XYZ/TMS tiling protocols or OGC-compliant Web Map Services (WMS/WMTS).

4.5 External Interface Requirements

- **User Interfaces:** Web Application responsive across desktop and tablet screen sizes (\$1920\times1080\$ down to \$768\times1024\$).
- **Software Interfaces:**
 - NASA Earthdata / CMR API (REST/HTTPS)
 - NOAA Big Data Program (AWS S3 Public Buckets)
 - OGC Web Standards (WMS, WFS, WCS)
- **Hardware / Network:** Minimum server network bandwidth \$1\text{ Gbps}\$ uplink for handling raster tile delivery; Client browser requiring WebGL support for vector and raster rendering.

4.6 Detailed Use Case Descriptions

Use Case: UC-02 — Perform Time-Series Trend Analysis

- **Primary Actor:** Researcher / Analyst
- **Preconditions:** Satellite dataset layers are processed, indexed, and available in object storage.
- **Main Success Scenario:**
 - User navigates to the interactive map interface and selects a parameter layer (e.g., Rainfall).
 - User defines an Area of Interest (AOI) by drawing a bounding box or selecting

an administrative boundary.

- User sets a historical time window (e.g., Jan 2020 – Jan 2025).
 - User clicks "Generate Time-Series Analysis".
 - System queries stored spatial rasters, calculates mean/median values per time step over the AOI, and renders an interactive line graph with trendlines.
 - User hover-inspects data points or exports calculated values as CSV/JSON.
- **Alternative Flow:**
 - 4a. Area of Interest exceeds spatial limits ($> 100,000 \text{ km}^2$): System prompts user to use reduced temporal resolution or select a smaller region to prevent server timeout.

5. Assumptions and Constraints

Assumptions

1. **Data Availability:** NASA, NOAA, and Copernicus public APIs maintain free access endpoints with acceptable uptime and rate limits.
2. **Standards Compliance:** Input satellite data includes standardized spatial coordinate reference systems (CRS) like EPSG:4326 or EPSG:3857.
3. **Client Capabilities:** End-user web browsers support modern HTML5, WebGL, and JavaScript ES6 features.

Constraints

1. **Technical:** High-resolution satellite raster processing is computationally heavy; backend raster routines must utilize parallel chunking (e.g., Dask, GDAL) to prevent latency spikes.
2. **Operational:** Automated scrapers/ingestion pipes must respect remote API rate limits (e.g., max 10 requests/sec per API key).
3. **Economic:** Cloud storage costs must be optimized by setting lifecycle management policies for older, low-demand cache tiles.
4. **Legal:** Compliance with open data policies and proper attribution to data source providers (NASA, NOAA, ESA).

6. Requirement Validation & Verification

Completeness & Consistency Check

Checked Area	Finding / Risk	Enhancement / Correction Strategy
API Rate Limits	High volume ingestion from NASA/NOAA may lead to IP throttling during major weather events.	Implemented an asynchronous message queue (RabbitMQ/Celery) with rate-limit aware workers and local caching.
Processing Latency	Computing statistical anomalies on multi-gigabyte rasters in real-time will freeze the UI.	Pre-compute baseline climatologies offline; calculate live anomalies as delta layers immediately upon file ingestion.
Bandwidth Overhead	Serving raw satellite rasters directly to browser clients exhausts user bandwidth.	Enforce Server-side Cloud-Optimized GeoTIFF (COG) windowing and XYZ web tile generation so clients load only visible pixels.
Missing Requirement	No specification for disconnected or poor network conditions in remote field usage.	Added Progressive Web App (PWA) client-side tile caching for small active AOIs.